Point estimation

POLI 210 - Seminar 7

Brad LeVeck
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The question behind an estimate

- We observe one study, but could have observed different outcomes.
- How does the rule producing our estimate behave across repeated studies?
- Does it target the right quantity? Does more data help?

Today: **point estimates → sampling distributions → causal comparisons.**

Optional extension: estimated standard errors and the t distribution.

Reading: PSD 5.6; a return to the CLT and t distribution from last week.

A motivating field experiment

Gerber, Green, and Larimer studied social-pressure mailings and turnout.

Mailing	Observed turnout
Neighbors	37.8%
Civic Duty	31.5%

The observed difference is about **6.3 percentage points**.

Is this the effect of the mailing, or could the groups differ for other reasons? **Random assignment is the key design feature.**

Example and rounded figures from Blackwell's estimation slides, based on Gerber, Green, and Larimer (2008). Civic Duty is an alternative mailing, not the no-mail control.

Estimand, estimator, estimate

Object	Meaning	Example
Estimand	Quantity we want	Average causal effect, τ
Estimator	Rule applied to random data	$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$
Estimate	Realized numerical result	About 6.3 percentage points

Before collecting data, the estimator is a **random variable**.

Its distribution depends on both the population and how we conduct the study.

What is a causal effect?

For unit i, imagine two potential outcomes:

$$Y_i(1): \text{ outcome under treatment}, \qquad Y_i(0): \text{ outcome under control}.$$

The individual effect is $\tau_i = Y_i(1) - Y_i(0)$.

For a fixed population of N units:

$$\tau = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i(1) - Y_i(0) \right].$$

We observe only one potential outcome for each unit, not its individual effect.

Assignment determines what we observe

If $D_i = 1$ means treated, the observed outcome is

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0).$$

Our estimate compares **different people**:

$$\hat{\tau} = \underbrace{\bar{Y}_1}_{\text{treated mean}} - \underbrace{\bar{Y}_0}_{\text{control mean}}.$$

A simulation can reveal both potential outcomes. An actual experiment cannot.

Assume well-defined treatments, no spillovers, and observation of assigned outcomes.

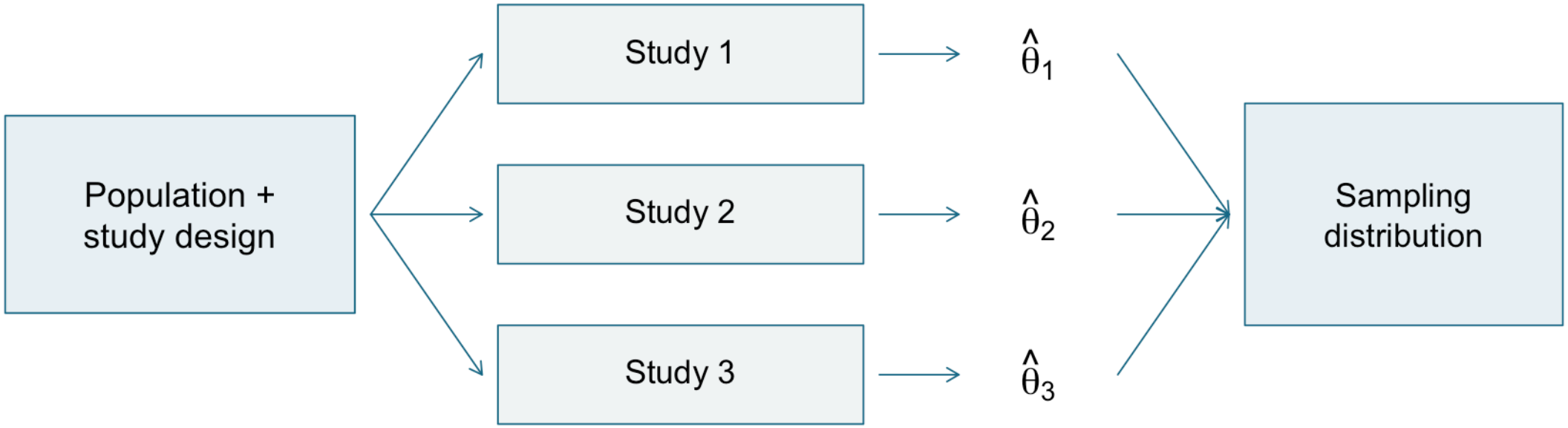
Two sources of randomness

Random assignment	Random sampling
Who receives treatment?	Who enters the study?
Can change while the units stay fixed	Produces a new set of units
Justifies treatment/control comparability	Connects a sample to a population

An experiment can use random assignment without a random population sample.

When describing a sampling distribution, ask: **What gets repeated?**

From repeated studies to a distribution



Each repetition produces **one estimate**, not one observation.

The sampling distribution is the distribution of that estimator.

A simple model for repeated turnout studies

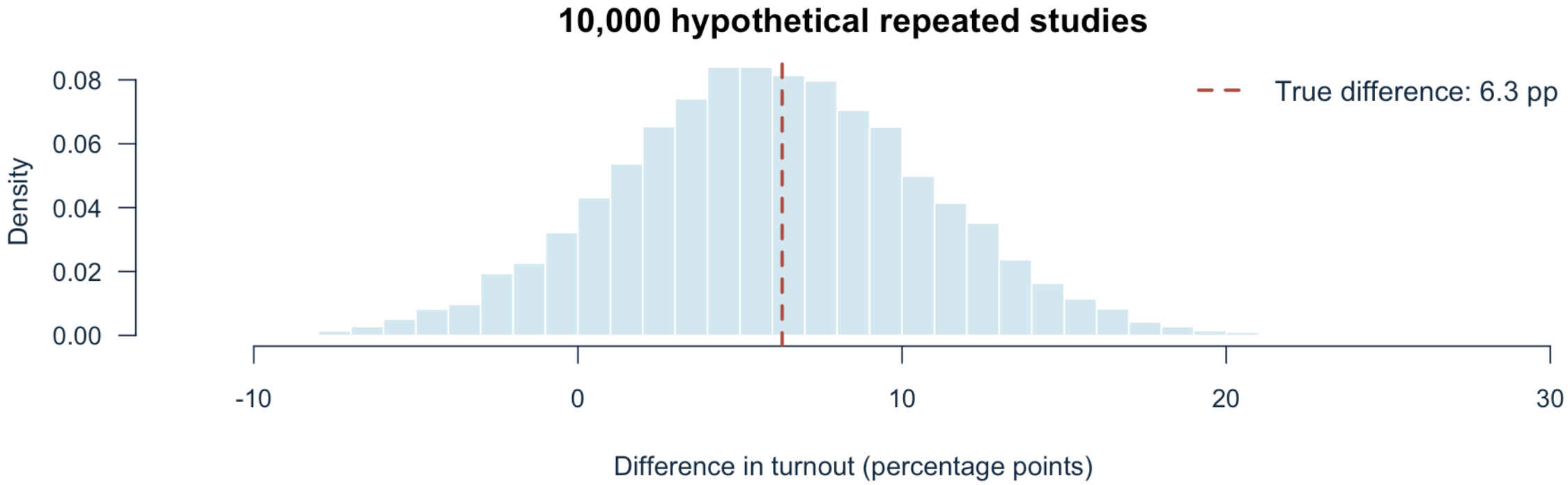
For illustration, suppose two independent samples of 200 voters have turnout probabilities 0.378 and 0.315.

```
set.seed(210071)
B <- 10000
estimates <- rbinom(B, 200, .378) / 200 -
             rbinom(B, 200, .315) / 200
```

This model's true difference is 0.063.

These are **invented repeated studies**, not a reconstruction of the original experiment's household assignment or standard errors.

One estimator, many possible estimates



The spread measures uncertainty across studies; the center addresses bias.

Unbiasedness: correct on average

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

An estimator is **unbiased** if $E[\hat{\theta}] = \theta$.

Under complete random assignment of a fixed number to each arm:

$$E_D[\bar{Y}_1] = \overline{Y(1)}, \qquad E_D[\bar{Y}_0] = \overline{Y(0)},$$

$$E_D[\hat{\tau}] = \overline{Y(1)} - \overline{Y(0)} = \tau.$$

The expectation averages over **assignments**, holding potential outcomes fixed.

Unbiased does not mean exact

- Each particular assignment can produce imbalance.
- Some estimates are too high; others are too low.
- Unbiasedness describes their average, not a guarantee for your study.

In simulation: compare `mean(estimates)` with the known truth.

In a small experiment: enumerate every possible assignment for an exact expectation.

Precision and standard error

The **standard error** is the SD of an estimator's sampling distribution:

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}.$$

For two **independent IID samples**:

$$\mathrm{Var}(\bar{Y}_1 - \bar{Y}_0) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

This is the model on slide 9. In a fixed pool allocated without replacement, the arm means are not independent; the variance calculation is different.

Mean squared error combines bias and spread

$$\mathrm{MSE}(\hat{\theta}) = E\left[(\hat{\theta} - \theta)^2\right] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

- Variance: spread around the estimator's own expectation.
- Squared bias: how far that expectation is from the target.

A very precise estimate can still be systematically wrong.

In simulation: `mean((estimates - truth)^2)`.

Consistency: concentration as n grows

$\hat{\theta}_n$ is **consistent** for θ if, for every $\epsilon > 0$,

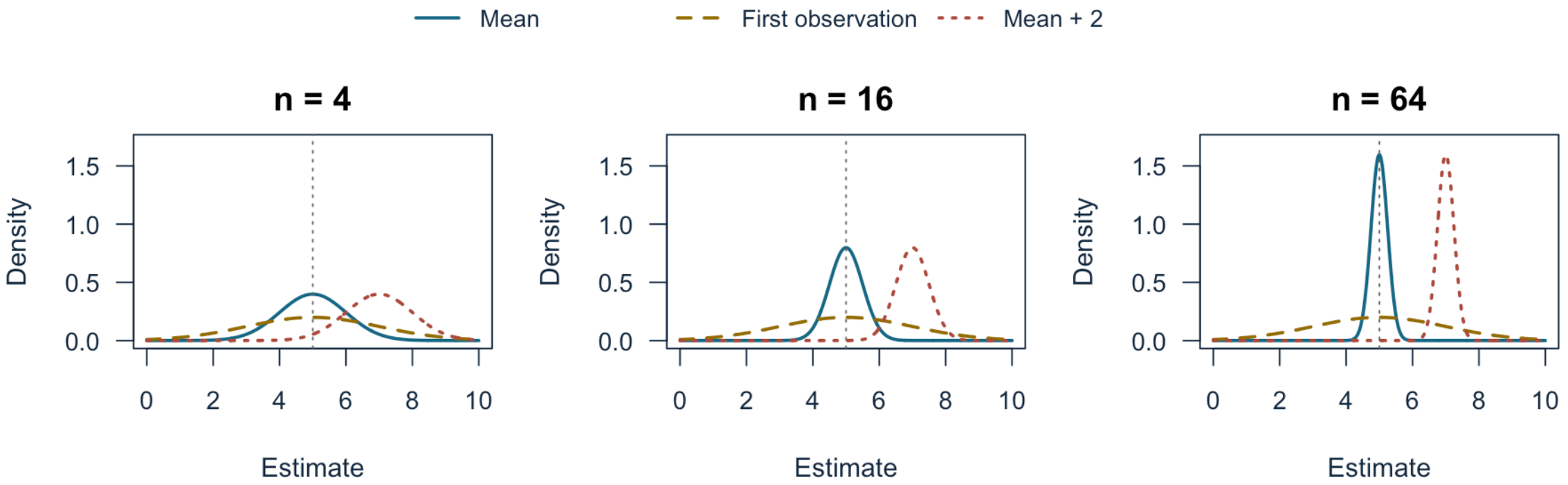
$$P\left(|\hat{\theta}_n - \theta| > \epsilon\right) \to 0.$$

Equivalently, $\hat{\theta}_n \xrightarrow{p} \theta$.

This is about a **sequence of larger studies**, not more repetitions of the same-sized study.

Unbiasedness asks about the center at a given n; consistency asks about concentration as n grows.

Centering and concentration are different



Here $X_i \overset{\text{iid}}{\sim} N(5, 2^2)$. Which estimator is unbiased? Which is consistent for 5?

Unbiasedness is neither enough nor necessary

Not enough: X_1 estimates μ without bias, but ignores additional data. Its spread does not shrink.

Not necessary: for IID data with variance σ^2,

$$V_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad E[V_n] = \frac{n-1}{n}\sigma^2.$$

V_n is biased downward at finite n, yet $V_n \xrightarrow{p} \sigma^2$.

Finite-sample bias can vanish as the sample grows.

A useful route to consistency

Apply Markov's inequality to $(\hat{\theta}_n - \theta)^2$:

$$P(|\hat{\theta}_n - \theta| \geq \epsilon) \leq \frac{E[(\hat{\theta}_n - \theta)^2]}{\epsilon^2} = \frac{\mathrm{MSE}(\hat{\theta}_n)}{\epsilon^2}.$$

So **MSE going to zero is sufficient** for consistency.

For $\bar{X}_n$, bias is zero and variance is σ^2/n. This recovers our finite-variance proof of the weak LLN.

More data must address the right problem

For independent samples, both group sizes must grow to drive

$$\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0} \to 0.$$

But shrinking this variance does not remove a persistent difference between **the comparison we estimate** and **the causal effect we want**.

We'll return to this distinction with self-selection after the break.

Break

When we return:

1. Why large observational studies can miss a causal effect.
2. Work through potential outcomes and the B3 simulation workflow.
3. If time permits: estimated standard errors and the t distribution.

Return to the causal question

In an observational study, people may choose treatment.

The comparison we observe is

$$E[Y \mid D = 1] - E[Y \mid D = 0] = E[Y(1) \mid D = 1] - E[Y(0) \mid D = 0].$$

But the population ATE is $E[Y(1) - Y(0)]$.

Why might these differ even with a huge sample?

Add and subtract the missing comparison

Add and subtract $E[Y(0) \mid D = 1]$:

$$\begin{aligned} E[Y \mid D = 1] - E[Y \mid D = 0] &= E[Y(1) \mid D = 1] - E[Y(0) \mid D = 1] \\ &\quad + E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0] \\ &= \text{ATT} + \text{baseline selection} \ . \end{aligned}$$

ATT is the average treatment effect among the treated.

The treated and control groups may differ **even without treatment**.

ATT is not automatically ATE

$$\text{Observed population comparison} - \text{ATE} = (\text{ATT} - \text{ATE}) + \text{baseline selection}\ .$$

- **ATT:** average effect for those who choose treatment.
- **ATE:** average effect for the entire target population.
- People may select on baseline outcomes, treatment effects, or both.

Population-level random assignment makes treatment independent of potential outcomes. In a finite experiment, the corresponding equalities hold **in expectation over assignments**, not for every realized allocation.

Build the observed outcomes in R

A different four-unit toy example for the live demonstration:

```r
y0 <- c(10, 20, 15, 25)
y1 <- c(15, 22, 21, 25)
D <- as.integer((1:4) %in% c(1, 3))
y <- ifelse(D == 1, y1, y0)
mean(y[D == 1]) - mean(y[D == 0])
```

`%in%` identifies treated units; `ifelse()` chooses which outcome we observe. Square brackets select a group. Only `y` and `D` enter the estimator.

Two B3 thought experiments

First: fixed seven villages	Later: larger studies
Keep potential outcomes fixed	Draw new villages from seven types
Vary which two are treated	Compare random assignment and self-selection
Enumerate $\binom{7}{2} = 21$ assignments	Simulate studies at several n
Study unbiasedness exactly	Explore concentration and limiting targets

n: villages per study. **B:** simulated repetitions.

Increasing B sharpens the simulation; it does not fix the study design.

Summarize estimates, not raw outcomes

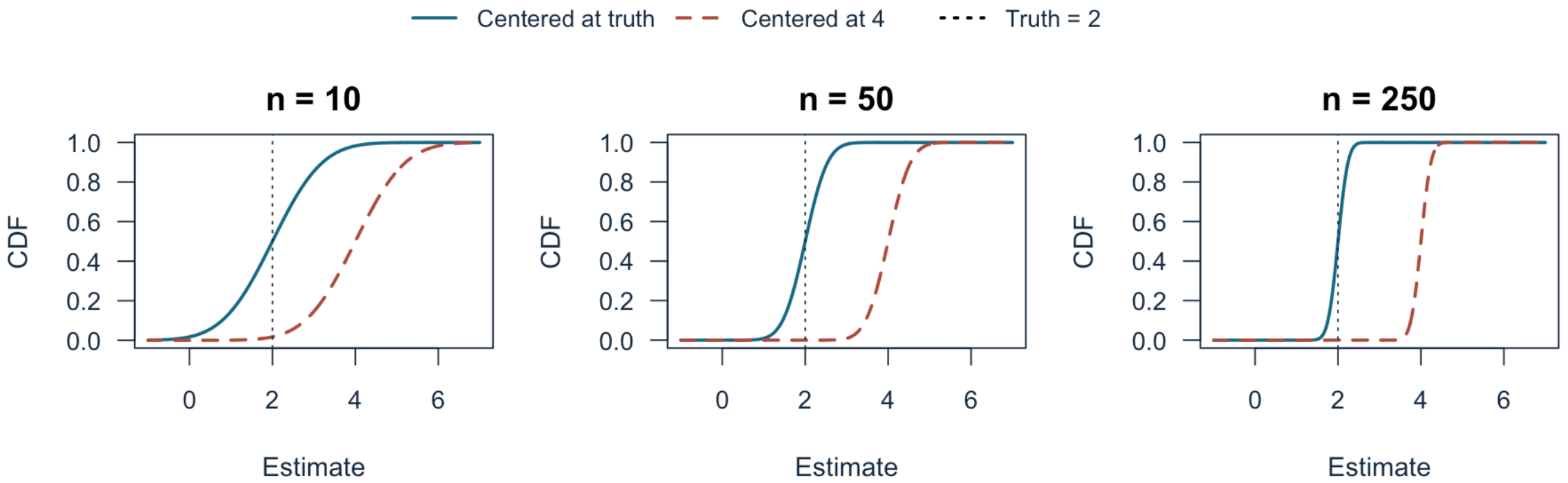
For `estimates`, a vector of B repeated estimates:

```
mean(estimates) - truth                # simulated bias
sd(estimates)                         # sampling SD
mean((estimates - truth)^2)            # MSE
mean(abs(estimates - truth) <= 1)      # within 1 point
```

`sd(estimates)` estimates the estimator's standard error.

Dividing by $\sqrt{B}$ instead estimates the Monte Carlo uncertainty in the **average simulated estimate**. It is a different quantity.

Read concentration from a CDF



A narrow range containing most estimates creates a **steep CDF transition**. Where is the transition? Steeper does not necessarily mean more accurate.

Takeaways and B3

- Unbiasedness concerns an expectation; consistency concerns concentration.
- Random assignment supports causal comparisons; sample size alone does not.
- Increasing n and increasing B answer different questions.

B3: one hypothetical village example, supplied simulation helpers, and short explanations. Use MSE and CDFs to distinguish precision from accuracy.

Retrieval question: Can an estimator become increasingly precise while remaining inconsistent for the ATE? Explain without using a formula.

Optional extension: estimated SEs and t

How does estimating σ change the distribution of a standardized mean?

An extension to revisit when we study statistical inference.

SD of observations versus SE of a mean

For IID observations:

$$\mathrm{SD}(X_i) = \sigma, \qquad \mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Usually σ is unknown. Estimate it with the sample SD:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \widehat{\mathrm{SE}}(\bar{X}) = \frac{S}{\sqrt{n}}.$$

R: `sd(x)` estimates the spread of observations; `sd(x) / sqrt(length(x))` estimates the SE of their mean.

Why divide by n − 1?

Estimating the center from the data makes squared deviations smaller:

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations for IID data with finite variance:

$$E\left[\sum_i (X_i - \bar{X})^2\right] = n\sigma^2 - n\frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Thus $E[S^2] = \sigma^2$. The sample variance is unbiased. This does **not** imply $E[S] = \sigma$.

Why does the estimated SD settle down?

Write the sample variance in terms of two averages:

$$S^2 = \frac{n}{n-1}\left(\frac{1}{n}\sum_i X_i^2 - \bar{X}^2\right).$$

For IID data with finite second moment, the LLN gives

$$S^2 \xrightarrow{p} E[X^2] - \mu^2 = \sigma^2.$$

If $\sigma > 0$, then $S \xrightarrow{p} \sigma$. **Consistency**, rather than unbiasedness, is what we need next.

Studentizing the sample mean

$$T_n = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \underbrace{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}_{\xrightarrow{d} N(0,1)} \underbrace{\frac{\sigma}{S}}_{\xrightarrow{p} 1} \xrightarrow{d} N(0,1).$$

This is an application of **Slutsky's theorem**.

We need IID observations with $0 < \sigma^2 < \infty$. The estimated SE becomes correct **relative to** the true SE:

$$\frac{S/\sqrt{n}}{\sigma/\sqrt{n}} = \frac{S}{\sigma} \xrightarrow{p} 1.$$

A finite-sample answer under Normal sampling

If $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, then

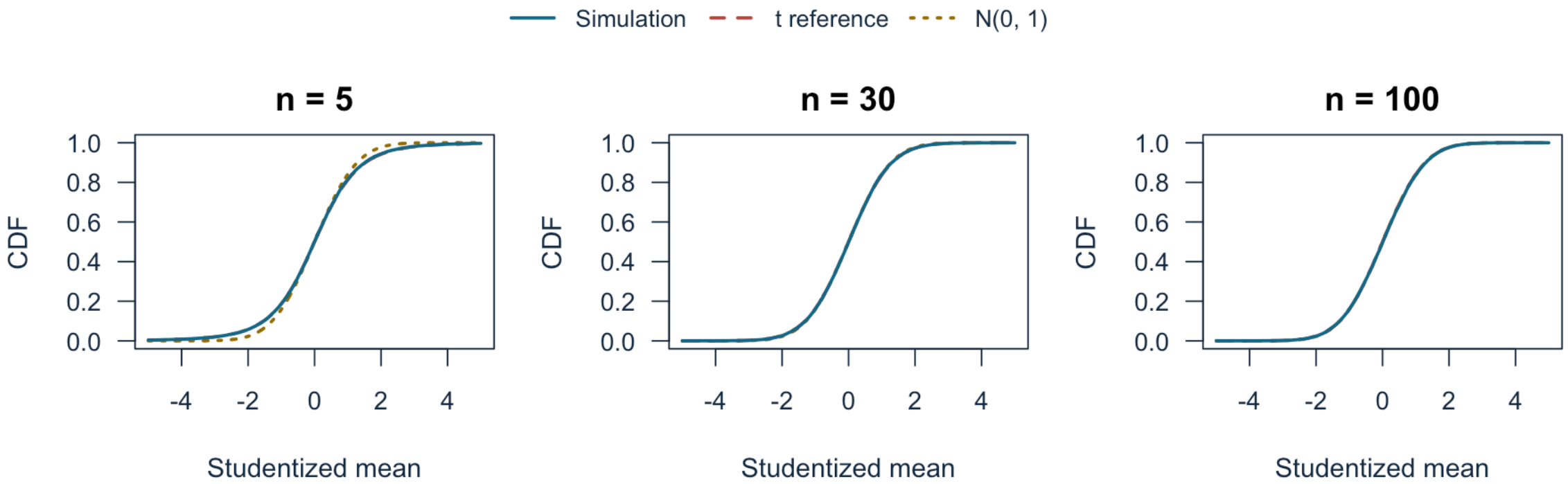
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1), \qquad U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

and Z and U are independent. Therefore

$$T_n = \frac{Z}{\sqrt{U/(n-1)}} \sim t_{n-1} \quad \textbf{exactly} .$$

The random denominator reflects **estimating the variance**, not an arbitrary way to add spread. Small estimated SDs can produce large ratios.

Exact t: compare the CDFs



Normal observations: $\mu = 5$, $\sigma = 5$; 20,000 repetitions per n. Simulation estimates the CDF; the t result itself is exact.

Make a small tail difference visible

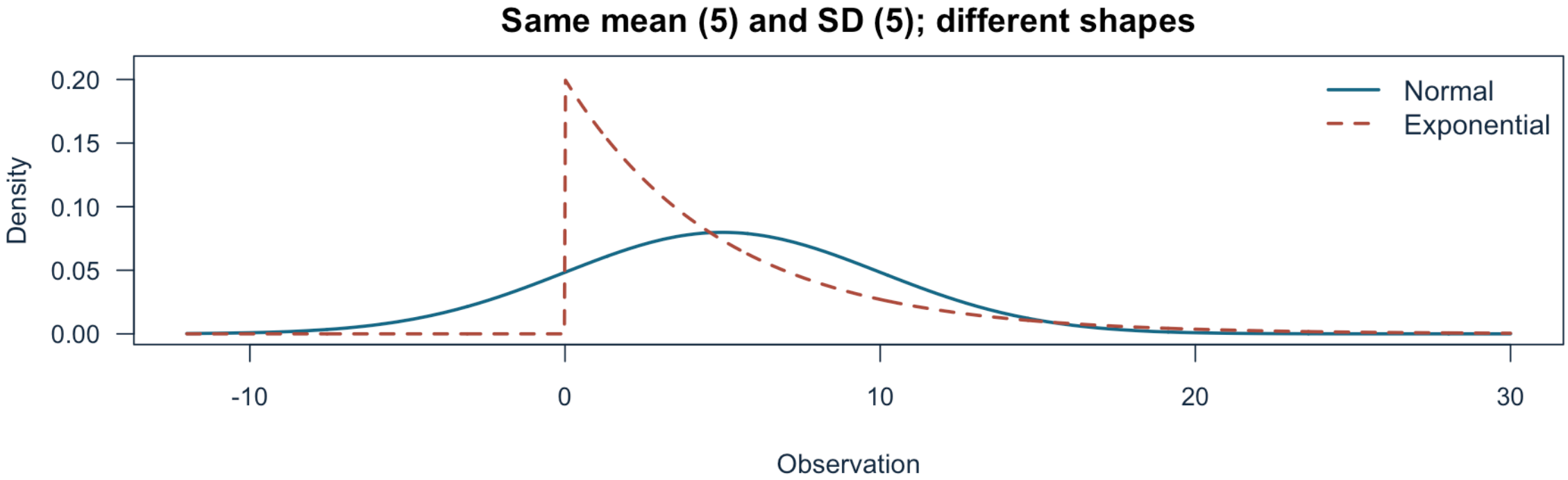
For Normal observations, compare $P(T_n < -2)$ and $P(T_n > 2)$.

n	Left (sim.)	Right (sim.)	Each t tail	Each Normal tail
5	0.058	0.056	0.058	0.023
30	0.026	0.027	0.027	0.023
100	0.023	0.024	0.024	0.023

At $n = 5$, t_4 has variance $4/(4-2) = 2$, not 1. The t variance is $\nu/(\nu-2)$ when $\nu > 2$.

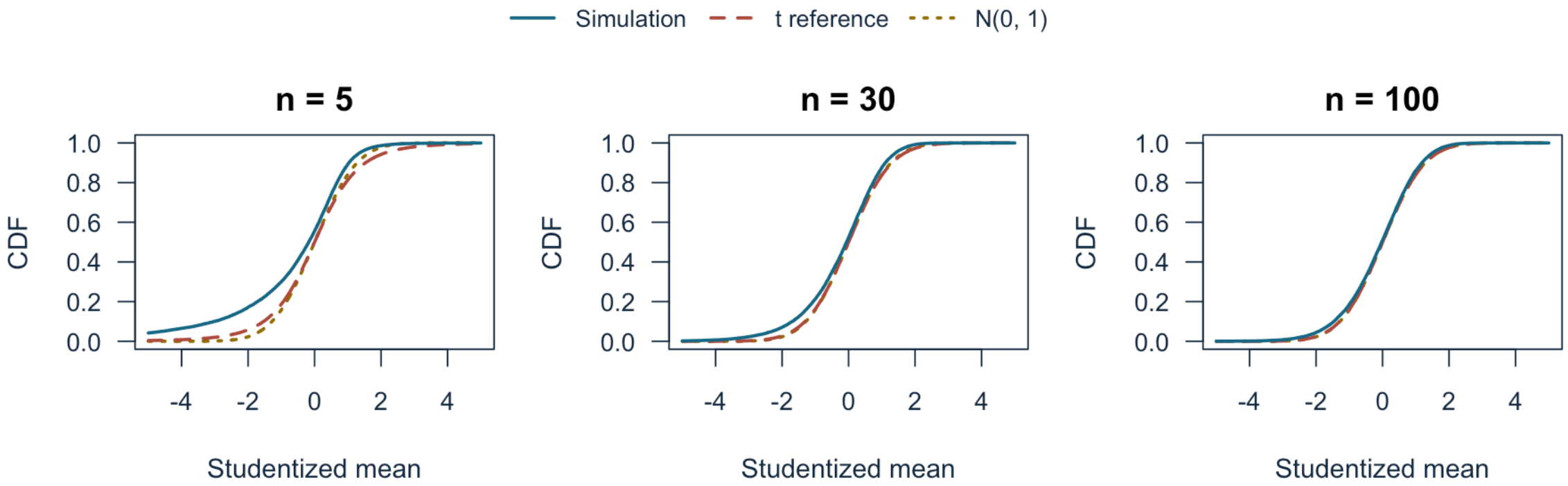
These are **event probabilities**, not hypothesis tests yet.

Change the population, keep its moments



Same μ and σ does not mean the same small-sample behavior. Use `rexp(n, rate = 1/5)` instead of `rnorm(n, 5, 5)`.

Studentized does not automatically mean t



Exponential observations: the small-sample distribution is **not exactly t**. The estimated denominator and numerator can move together asymmetrically.

Check both tails, not just total spread

n	Left (sim.)	Right (sim.)	Each t tail	Each Normal tail
5	0.172	0.014	0.058	0.023
30	0.071	0.008	0.027	0.023
100	0.045	0.013	0.024	0.023

- A symmetric t curve cannot reproduce this finite-sample asymmetry.
- Increasing n improves the Normal approximation here, but not instantly.
- There is no universal sample-size cutoff guaranteeing a good approximation.

The lesson we will reuse for inference

Claim	What supports it?
$T_n \sim t_{n-1}$ exactly	IID Normal observations
$T_n \xrightarrow{d} N(0,1)$	IID, finite positive variance
A reference curve works at a particular n	Assumptions and approximation quality

In the live demo, compare `ecdf(t_values)` with `pt()` and `pnorm()`.

Confidence intervals and tests will use these ideas later.

Reading and references

- Speegle and Clair, *Probability, Statistics, and Data*, Section 5.6; revisit the t distribution in Section 5.5.
- Matthew Blackwell, Estimation and Inference slides, Sections 1-2. Instructional structure and turnout example adapted with permission.
- Gerber, Green, and Larimer (2008), “Social Pressure and Voter Turnout,” *American Political Science Review* 102(1): 33-48.
- Gerber and Green (2012), *Field Experiments*, Chapter 2, especially Table 2.1 and Section 2.6. Potential outcomes and selection-bias decomposition.
- R documentation: `TDist`.

Figures and simulation code were created for this seminar. The village and four-unit examples are hypothetical, not empirical effect estimates.